

Project Executable Files

Date	02 july 2026
Team ID	SWTID-2026-2152
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	3 Marks

Project Executable Files

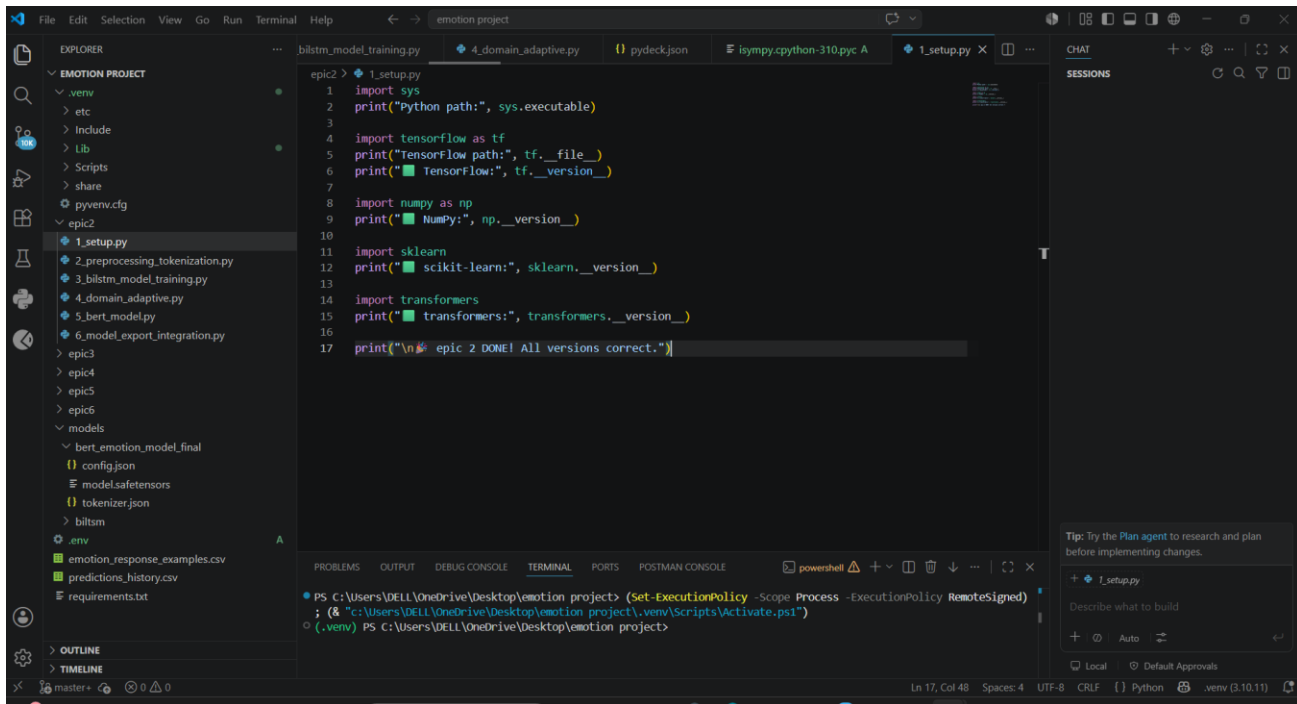
This section requires submission of all executable and deployable files for your project. Ensure all files are properly organized, named, and uploaded to the specified submission platform. The evaluator must be able to run or deploy your solution using the provided files and instructions.

Step 1: Submission Checklist

S.No	Item to Submit	What to put / How to decide
1	Complete source code (all files and folders)	Yes (This is mandatory for your machine learning/web code).
2	README / Setup Guide (instructions to run the project)	Yes (Include a .md or .txt file explaining how to install and run your engine).
3	requirements.txt / package.json / dependency file	Yes (If python-based, provide requirements.txt; if Node.js/React, provide package.json).
4	Database schema / seed files (if applicable)	Yes (If your project saves data, user profiles, or logs) or NA (if it runs entirely without an external database).
5	Environment configuration file (.env.example or similar)	Yes (If you use API keys or environment variables) or NA (if no hidden configuration keys are used).
6	Deployed application URL (if hosted)	Yes (If you hosted it on Render, Vercel, AWS, or Heroku) or NA (if it only runs locally).
7	APK / Executable binary (if applicable for mobile/desktop apps)	NA (Unless you built a specific .apk file for Android or .exe for Windows).
8	Dockerfile / Containerization config (if applicable)	NA (Unless you explicitly dockerized your application).
9	Test files and test results	Yes (If you wrote test scripts/validation logs for your emotion models) or NA .
10	Demo video or walkthrough (if required)	Yes (Highly recommended for project evaluations to show how the engine detects emotions).

Step 2: File / Folder Structure

Provide an overview of your project's file and folder structure below:



Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	https://emotion-detection-learning-support-engine-6sljxmpjdmsilahagvxj.streamlit.app/
Login Credentials (Demo)	Not Required
Platform / Hosting Provider	Streamlit Cloud
Repository Link	https://github.com/Amruthav421/Emotion-Detection-Learning-Support-Engine
Demo Video Link	https://drive.google.com/file/d/1tToMFLqie_9UxoHAWY-dZg-OKXEIlgNuR/view?usp=drivesdk

Step 4: Run Instructions:

Provide clear, step-by-step instructions for the evaluator to run your project locally.

Pre-requisites:

Python 3.10 or above

Visual Studio Code

Git

Internet Connection

Google Gemini API Key

Steps:

Step 1:

Clone the repository

`git clone <Repository URL>`

Step 2:

Open the project folder

`cd Emotion-Detection-Learning-Support-Engine`

Step 3:

Create and activate virtual environment

`python -m venv .venv`

`.venv\Scripts\activate`

Step 4:

Install dependencies

`pip install -r requirements.txt`

Step 5:

Create a .env file and add your Gemini API key

`GEMINI_API_KEY=your_api_key_here`

Step 6:

Run the application

`streamlit run app.py`

Step 7:

Open the browser

`http://localhost:8501` □

Step 8:

Enter student text or learning-related input in the application.

Step 9:

Click the Analyze Emotion button.

Step 10:

View the detected emotion, confidence scores, personalized learning support response, emotion visualization charts, and prediction history.

Step 11:

Save or download the generated results for evaluation and testing purposes.

Step 5: Known Issues / Limitations

S.No	Known Issue	Workaround / Status
1	Emotion prediction accuracy may vary for short or unclear text inputs	Provide meaningful and complete text for better results
2	Model supports only emotions present in the training datasets	Future versions will support more emotion categories
3	Performance depends on dataset quality and preprocessing	Additional datasets can improve accuracy
4	Sarcasm and highly contextual emotions may not always be detected correctly	Future versions may use advanced transformer models
5	Requires internet connection for Gemini API-based features	Ensure stable internet connectivity
6	Mixed-language (Telugu-English) inputs may reduce prediction accuracy	Future version will include multilingual support
7	Large text inputs may take longer to process	Keep input length within reasonable limits

Project Future Enhancements

S.No	Enhancement
1	Real-time emotion detection from voice input
2	Multilingual emotion analysis (Telugu, Hindi, English)
3	Personalized learning recommendations based on emotion
4	Student progress tracking dashboard
5	Advanced BERT/Transformer-based emotion models

